Perception: Stereo Holds and Robot Pose

Ege Gursoy

30 October 2026

The `perception_challenge_best` branch contains the selected hold estimator, robot estimator and RGB robot/cup tracker. The command reference is `docs/perception/README.md`; historical alternatives remain on `perception_challenge_full`.

1 Input, output and geometry

The image challenge provides a stereo RGB pair. The estimator returns each hold's type, color, backplate-anchor position in metres and intrinsic XYZ Euler orientation in radians. The scene frame has its origin at the wall's bottom centre: x points right, y up and z out. A mesh centre is different from its backplate anchor. These conventions determine the submission coordinates.

The updated public source supports Cube, Microban, Sidepull and Fingerpocket. We pin the four meshes and scene geometry to source commit `51efe48`. The supplied sample rig has a 3 m by 1.4 m wall, a 64 mm stereo baseline, 1 m camera distance, 0.7 m camera height and 74° vertical field of view. Each stereo panel is 699 by 318 pixels. These are declared inputs; labelled hold positions are never used to estimate camera calibration.

Wall width also changes the tetrahedral volumes. Their base height is $\min(0.1w, h)$, where w, h are the wall dimensions. The supplied wall therefore uses a 0.3 m base height. The former 0.8 m wall and 0.08 m volume assumptions displaced inclined support faces. We now use the declared wall size throughout rendering, fitting and JSON export.

2 Selected hold estimator

Find and pair instances. Color components initialize hold proposals. Epipolar consistency and projection onto the known wall pair observations across cameras. The old 100 mm clearance merge and narrow-instance gate were removed: the new scenes contain closer instances and wider Sidepull holds. Proposals still depend on the released rendered appearance.

Fit all four meshes in both images. For each proposal, the estimator renders all four normalized CAD meshes at the three public in-plane yaw choices: zero and plus/minus 45°. Candidate support surfaces are the wall and image-estimated volume faces. MuJoCo renders their silhouettes and occlusions; local optimization fits the candidate to both views. Promising inclined candidates receive translation and pose refinement. Flat face regions in RGB estimate the volumes, with hold pixels excluded from face refinement.

Fingerpocket has weak silhouette differences between yaw choices. A second check compares stereo RGB correspondences predicted by each candidate's rendered depth. This is CAD hypothesis depth, not a depth-camera measurement. The check chooses between close silhouette fits only when the RGB evidence has a sufficient gap.

Recover a compatible scene. The previous ten-Microban generator search is removed. The replacement uses estimated geometry: align each volume-mounted backplate to its support face,

apply the source's approximately 1.73 mm clearance, and check quantized convex hulls for overlap. The weaker image fit moves tangentially to its support plane when a conflict occurs. Unresolved candidates are rejected. Diagnostics record all changes. This stage does not recover a random seed or claim exact source JSON.

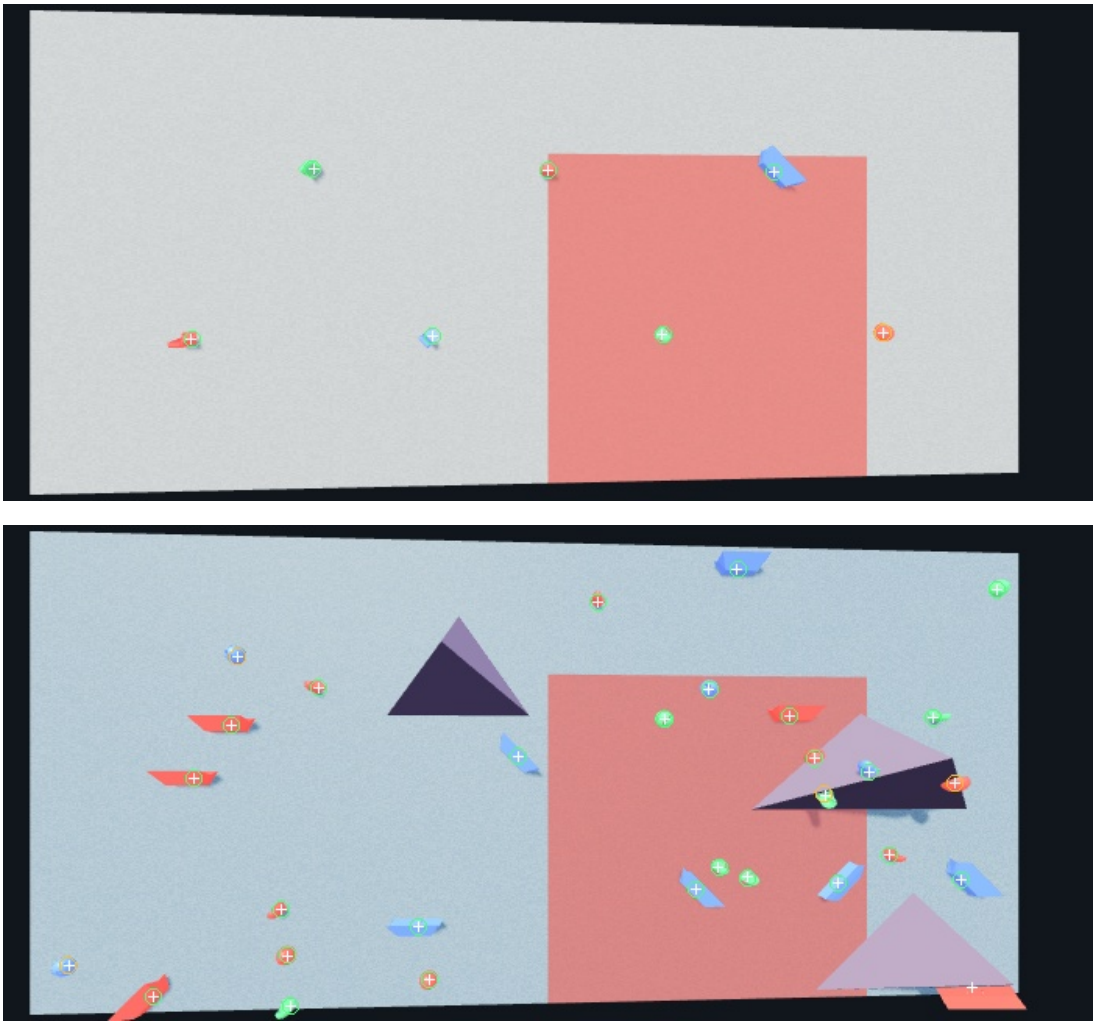


Figure 1: Left-view scoring overlays for the two retained fixtures: a controlled four-type scene including four Cubes (top), and a supplied three-volume scene (bottom). White crosses mark reference backplate anchors. Circles mark estimated anchors: green passes the local strict pose check; amber fails it. A correctly placed amber circle can indicate an orientation error. Inference uses both views; references are used only for scoring and this overlay.

3 Results and remaining errors

We evaluated all 40 supplied scenes, containing 1,200 holds. Matching requires the correct type and position error at most 50 mm. A strict pose additionally requires position error below 5 mm and rotation error below 15°, measured on SO(3). Missed holds remain in the denominator. These are local diagnostics; the repository does not supply an official weighted scorer.

Four scenes guided image-fit development. We froze the image-fitting files before evaluating the remaining 36: these gave 1,041/1,080 matches and 896 strict poses. The source-clearance correction was added after observing scene-validator failures on the full set. Table 1 therefore reports final regression results, not a fresh blind evaluation.

Table 1: All 40 supplied scenes, with 1,200 reference holds. Viewer acceptance checks scene compatibility and is separate from pose accuracy.

Stage	Matches	Strict	False positives	Viewer accepted
Frozen image fit	1,160	999	7	23/40
Support/hull recovery	1,159	1,005	8	40/40

The correction resolves volume overlaps and improves several inclined poses. It also moves one former match outside the position tolerance. We retain raw image predictions separately so this tradeoff remains inspectable.

Table 2: Final results by type in the supplied archives. Cube requires a separate control because it is absent from these archives.

Type	Reference holds	Matches	Strict poses
Microban	429	412	405
Sidepull	349	338	335
Fingerpocket	422	409	265
Cube	0	—	—

Orientation is the main remaining error. Among matched holds, 141 are within 5 mm but fail the rotation check. Table 2 shows that Fingerpocket accounts for most strict-pose failures. At this image resolution, its similar yaw appearances remain ambiguous after silhouette and RGB refinement. Thirteen matched holds exceed 5 mm position error. Partial visibility, missed components and uncertain volume faces also remain relevant; the current experiments do not isolate their individual contributions.

The supplied generator's random type list currently excludes Cube, despite supporting its mesh. We generated three additional controls by extending only that list. Their renderer, collision rules and volume geometry were unchanged. Across 48 holds, the estimator gives 47 matches and 37 strict poses; the 12 Cube instances give 11 matches and 10 strict poses. All three scene envelopes pass the upstream validator. These controls are separate from the supplied data.

What earlier trials established. The initial mixed-type pilot retained CAD fitting because it produced 51/60 matches and 45/60 strict poses on two scenes. None of its five volume-mounted holds passed the strict check, motivating the geometry update. The legacy sampler's assumptions were incompatible with the new source. Earlier FoundationPose, SAM-6D and stereo-network experiments remain on the full branch; they do not supply a 40-scene comparison against this updated estimator. Multiview CAD remains the deployed method. The new evaluation establishes its current performance, not superiority over unrun alternatives.

4 Run and inspect the implementation

Hold inference needs Linux x86-64, Python 3.12 and functioning EGL/OpenGL rendering. It does not require Torch or a CUDA compiler. Setup downloads the four pinned, checksum-verified meshes into a private cache.

```
python3.12 scripts/perception/setup.py
source .venv-perception/bin/activate
python scripts/perception/run.py doctor
python scripts/perception/run.py reproduce \
  --workers 2 --out outputs/perception/holds
```

The two retained fixtures contain 37 holds. A fresh environment and empty cache completed installation and reproduction: 36 matches, 31 strict poses, no false positives, and both exported scenes accepted by the upstream validator. A completed command does not imply that all poses passed.

For the supplied archives, use the declared sample rig:

```
python scripts/perception/prepare_samples.py \
  --samples samples --out data/local/perception_samples
python scripts/perception/run.py reproduce \
  --dataset data/local/perception_samples --workers 4 \
  --out outputs/perception/samples
```

For another rig, supply measured calibration. A new pair uses `holds -stereo stereo.png -calibration rig.json -out DIR`. GPU jobs wait for spare capacity and poll every 30 minutes while externally busy.

Table 3: Hold implementation under `src/rcr/perception/challenge/src/`.

File	Role
`hold_scene.py`	RGB proposals, four-type hypotheses and selection
`hold_geometry.py`	Wall dimensions, volume estimation and face refinement
`hold_fit.py`	Dual-view losses, local fitting and stereo RGB check
`hold_constraints.py`	Support clearance and convex-hull recovery
`rcr_exact_fit.py`	Explicit per-mesh CAD renderer
`rcr_contract.py`	Challenge JSON validation and local scoring

Outputs separate raw fits (`image_holds.json`), recovered poses (`holds.json`), the viewer scene (`prediction.json`) and diagnostics (`debug.json`). `docs/perception/benchmark.json` preserves per-scene summaries and freeze hashes. `report_figures.py -predictions DIR` rebuilds the overlays.

5 Robot pose and physical cameras

Robot implementations are unchanged. Saved-frame `max` uses Microban keypoints, measured-joint configured M3T and protected CAD/boundary refinement. Continuous `fast` adds persistent tracking, RGB cup mapping and conditional socket correction. Both require all 19 measured joints. Add `setup.py -robot` for their dependencies and `-cameras` for color-only RealSense capture.

The retained robot fixture passes root and endpoint checks: 0.879 mm translation, 0.297° rotation and 1.759 mm worst endpoint error. Combined replay returns a valid root pose and 72 cup candidates without a refresh error; candidate count alone is not map accuracy. Shared team types, submodules, planning and controller code are unchanged.

The guide gives capture and chessboard-calibration commands for two selected camera serials. Measure the square size and calibrate the actual resolution, serial order and capture profile. Estimate wall pose separately. Encoder acquisition times must share the camera host's monotonic clock. No physical cameras were connected: USB capture, exposure synchronization and physical accuracy remain unvalidated.

Keep challenge backplate JSON separate from the physical cup map and team bolt coordinates. Socket proximity never confirms loading. The next camera test must measure calibration and pose errors on the actual rig. The rendered-scene result supports the four-type CAD path; small Fingerpocket images still leave substantial orientation ambiguity.